

MEDXAI CLINICAL MRI RESEARCH REPORT

Report ID:	6c3db14f-979e-4d03-9421-58b9a5fc61e6
Date/Time:	2026-08-13 06:52:41.352818
Scan Filename:	OIP.jpg
Diagnosis Classification:	Non Demented
Model Confidence:	91.84%

AI Prediction Summary

The MEDXAI EfficientNetB0 model classified the submitted MRI image as **Non Demented** with a model confidence of **91.84%**. The model produced a high-confidence classification for the submitted MRI image.

Probability Analysis

The highest model probability is associated with **Non Demented** at **91.84%**. The next highest probability is **Mild Demented** at **7.71%**, giving a probability difference of **84.13 percentage points**. This indicates a comparatively strong separation between the leading class and the next most likely class.

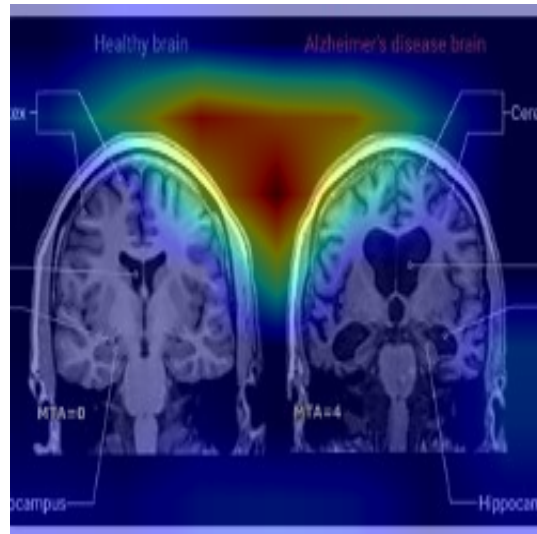
Class Probability Breakdown

Class	Probability
Non Demented	91.84%
Mild Demented	7.71%
Moderate Demented	0.45%
Very Mild Demented	0.00%

Original MRI Scan



Grad-CAM Explainability Heatmap



The Grad-CAM visualization provides a model-focused representation of the image regions that contributed most strongly to the **Non Demented** classification. Highlighted regions should be interpreted as areas receiving greater influence from the neural network rather than as definitive evidence of disease.

Overall AI Findings

The overall MEDXAI analysis classified the MRI as **Non Demented** with a confidence of **91.84%**. Grad-CAM was successfully generated, while a usable LIME visualization was not available. The probability distribution describes how the model allocated prediction likelihood across the available classes, while Grad-CAM and LIME provide complementary views of the model's decision process. These explainability methods describe model behavior and do not independently establish the presence, absence, severity, or stage of Alzheimer's disease.

- Leading classification: **Non Demented**
- Leading probability: **91.84%**
- Reported model confidence: **91.84%**
- Explainability interpretation: highlighted regions represent areas that influenced the AI model's decision and should not be treated as independently diagnostic.

MEDICAL DISCLAIMER: This AI-generated report is produced by the MEDXAI EfficientNetB0 Engine for research and clinical decision-support purposes. The prediction, probability values, Grad-CAM visualization, and LIME explanation describe the behavior of an artificial intelligence model and are not a medical diagnosis. This report must be reviewed by a qualified radiologist, neurologist, or other appropriately qualified healthcare professional before any clinical diagnosis or treatment decision is made.